



Università degli Studi di Napoli Federico II
Scuola Politecnica e delle Scienze di Base

Dipartimento di Ingegneria Industriale
Corso di Laurea in Ingegneria Meccanica

Classe di laurea L 09

Elaborato di laurea in

Tecnologia meccanica

Superalloys and casting process for ultra-high temperature gas turbines

Relatore:

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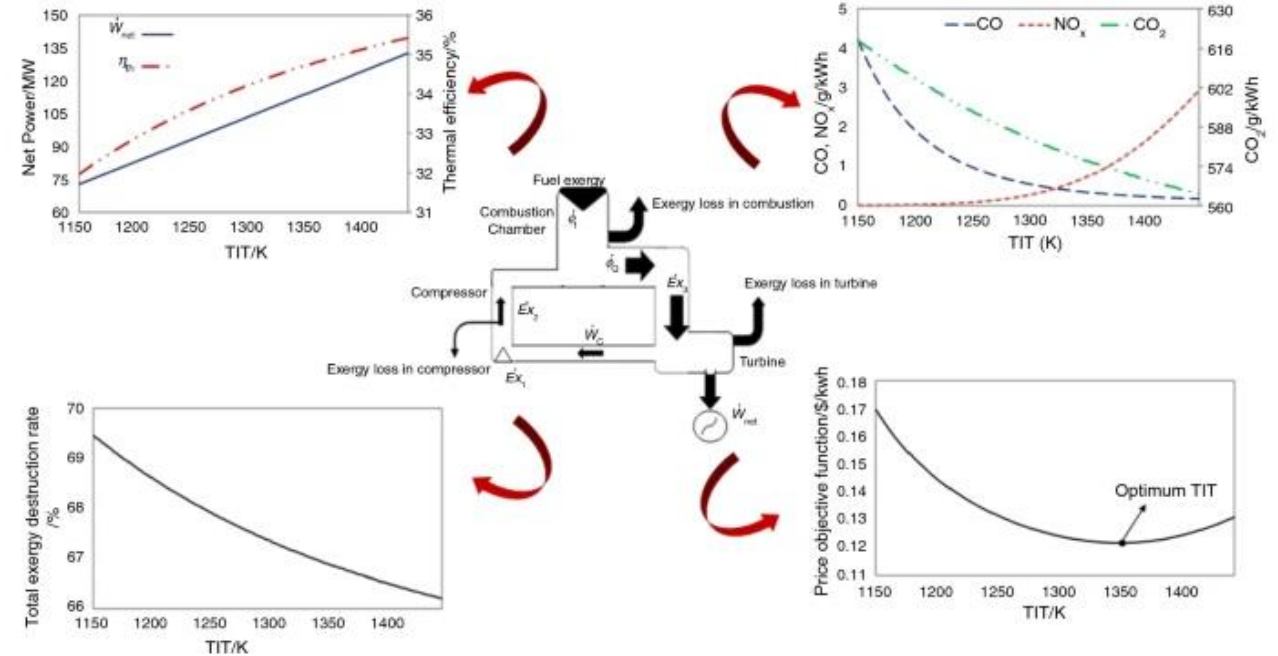
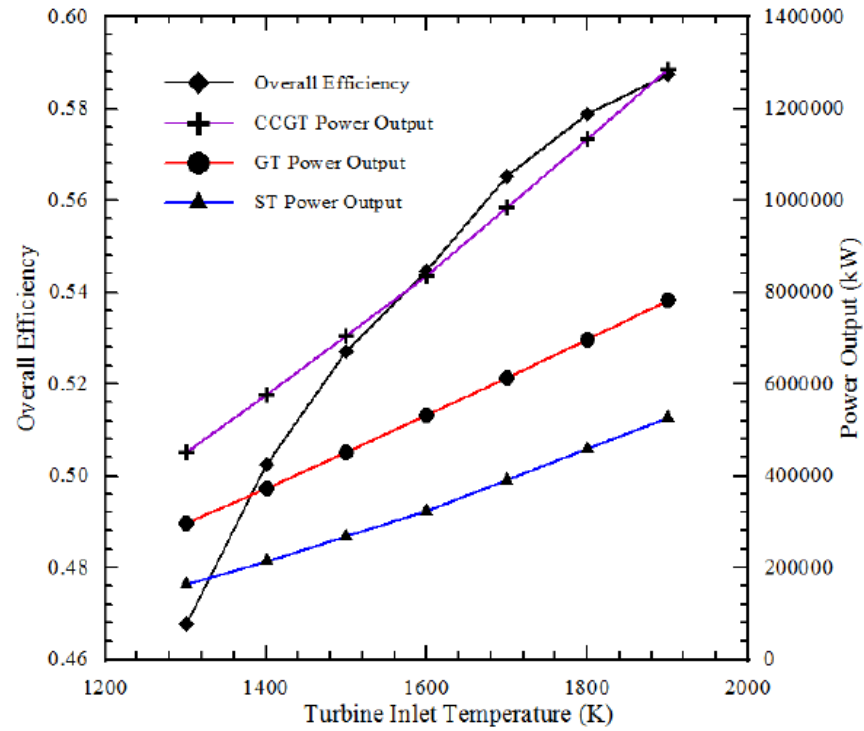
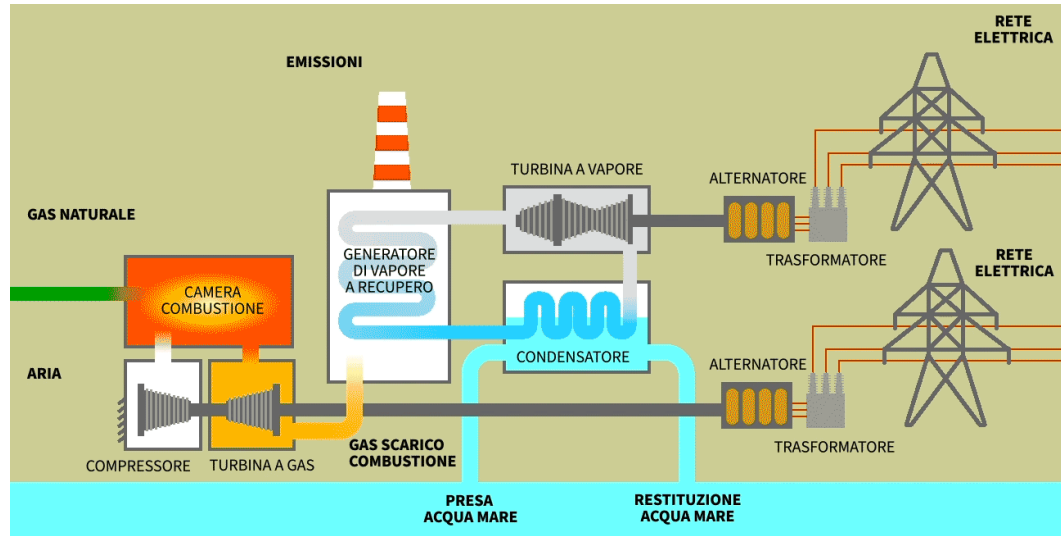
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Salvatore Palumbo

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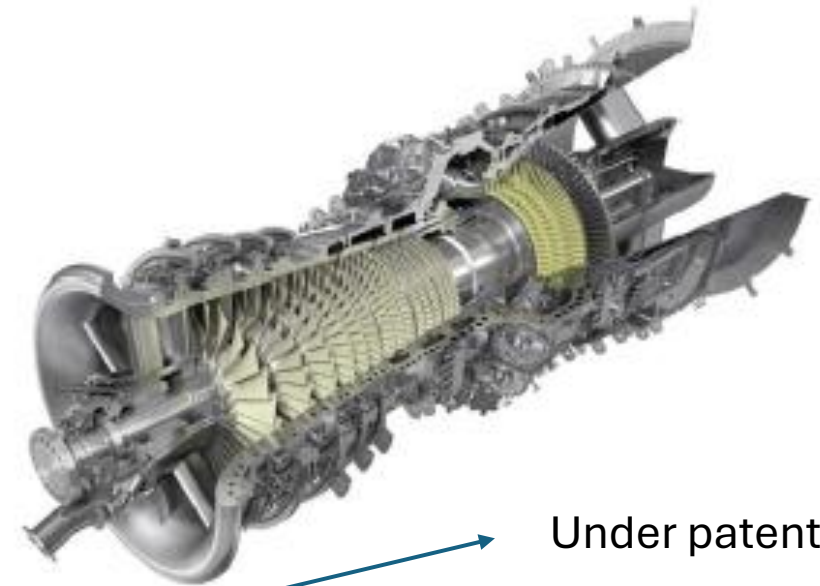
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Why increasing TIT?



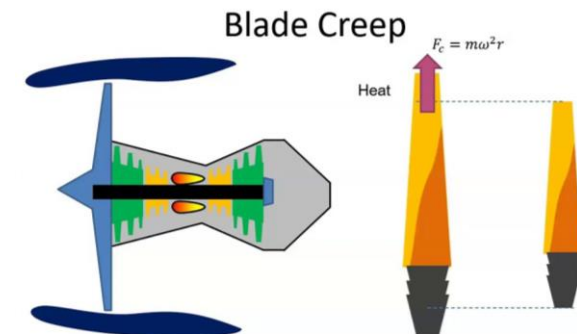
Mitsubishi Heavy Industries (MHI) project

- **High-performance compressors with improved boundary layer control** to increase pressure ratio and efficiency.
- **Lean premix combustors with reduced NOx emissions** achieved through improved flame position evaluation using optical measurement and large-scale unsteady numerical simulation.
- **High-performance turbines with optimized exhaust diffusers** to improve efficiency by reducing turbine blade exit velocity.
- **Advanced thermal barrier coatings (TBC) for high-temperature components** with improved thermal conductivity and high-temperature stability.
- **Welding repair technology for single-crystal turbine blades** using Laser Metal Deposition (LMD).
- **Additive manufacturing for complex cooling structures in hot parts** to reduce cooling air volume.
- **Super heat-resistant materials for turbine blades** with excellent creep strength and dwell thermo-mechanical fatigue strength.
- **Casting process simulations for directionally solidified and single crystal blades** to predict casting defects and optimize casting processes.
- **Wireless power supply technology for high-speed blade vibration measurement** eliminating the need for battery changes.
- **Blade internal surface defect detection technology** using a matrix array UT probe for improved inspection of cooling passages.



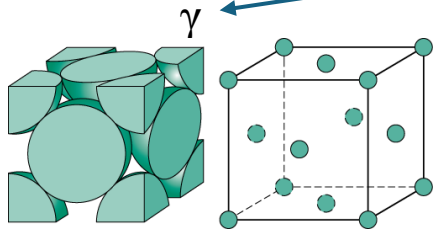
Under patent!!

Structural problem!!

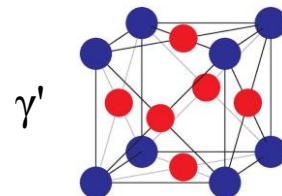
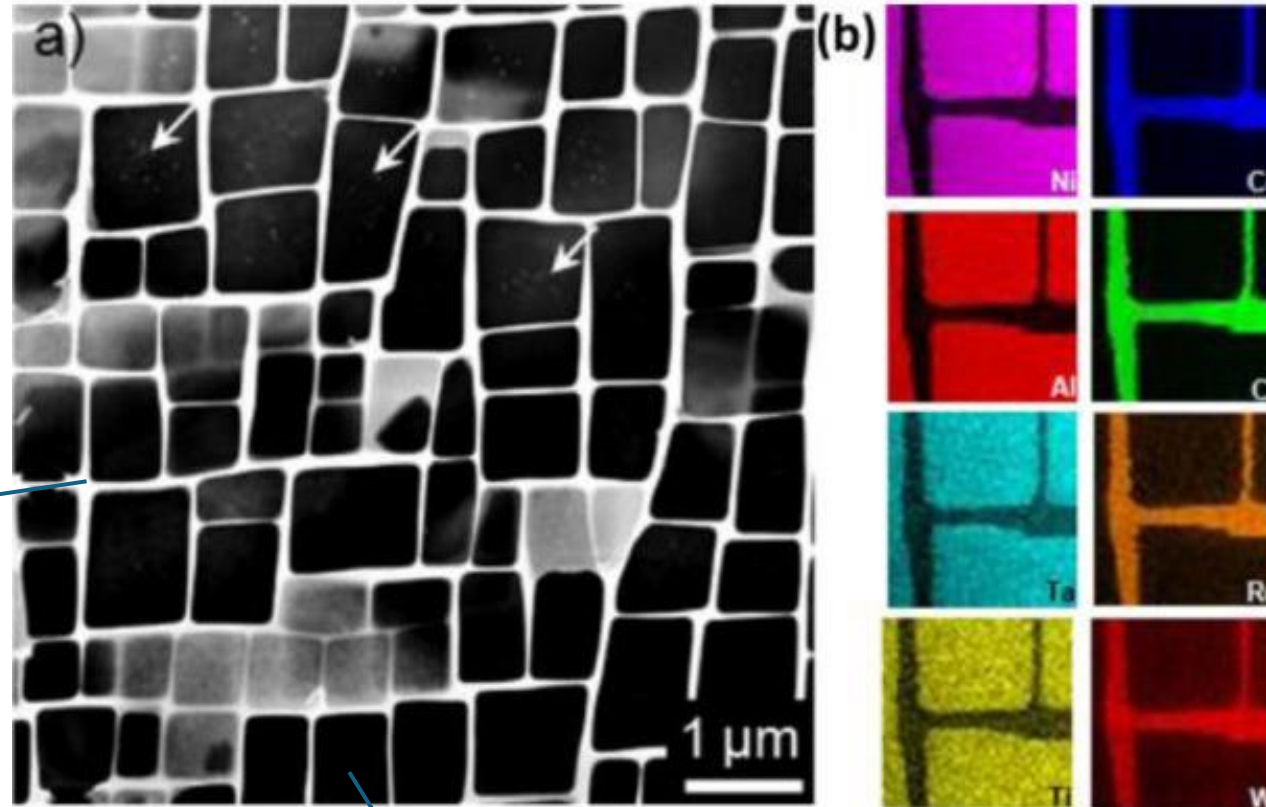


Ni-based superalloy microstructure and importance of volume fraction

Single Crystal
superalloys: a key
factor of the creep
strength



FCC structure



FCC-L1₂ structure
(ordered)

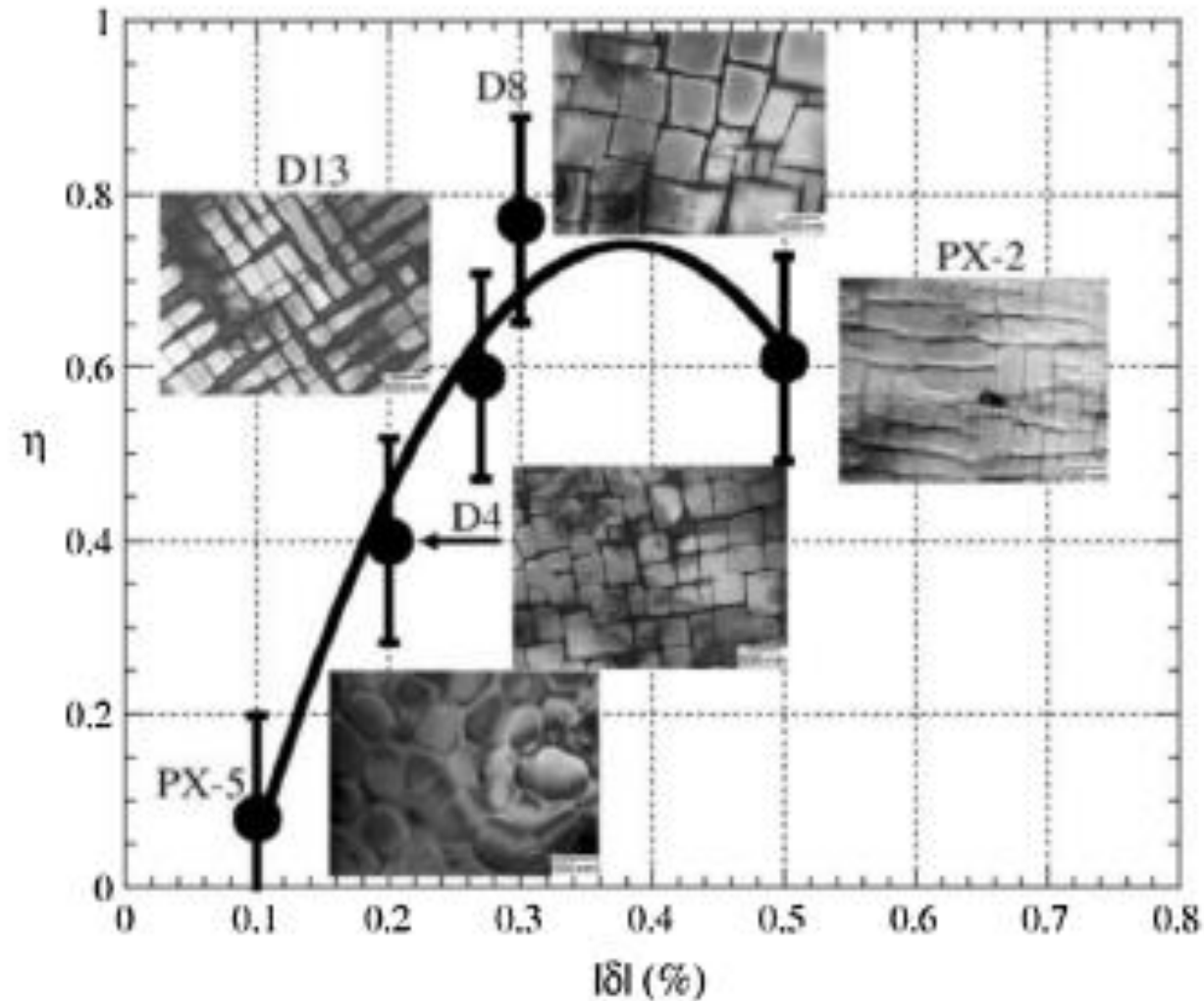
65%-70% in γ'



If more than this,
degradation of
cuboids at high
temperature!!

Lattice misfit:
a parameter to manage the γ/γ' interfacial coherency

$$\delta = \frac{(a_{\gamma'} - a_{\gamma})}{(a_{\gamma'} + a_{\gamma})}$$



Closely affect creep properties!!

Dislocation motion: a way to understand creep strength of Ni-based superalloy

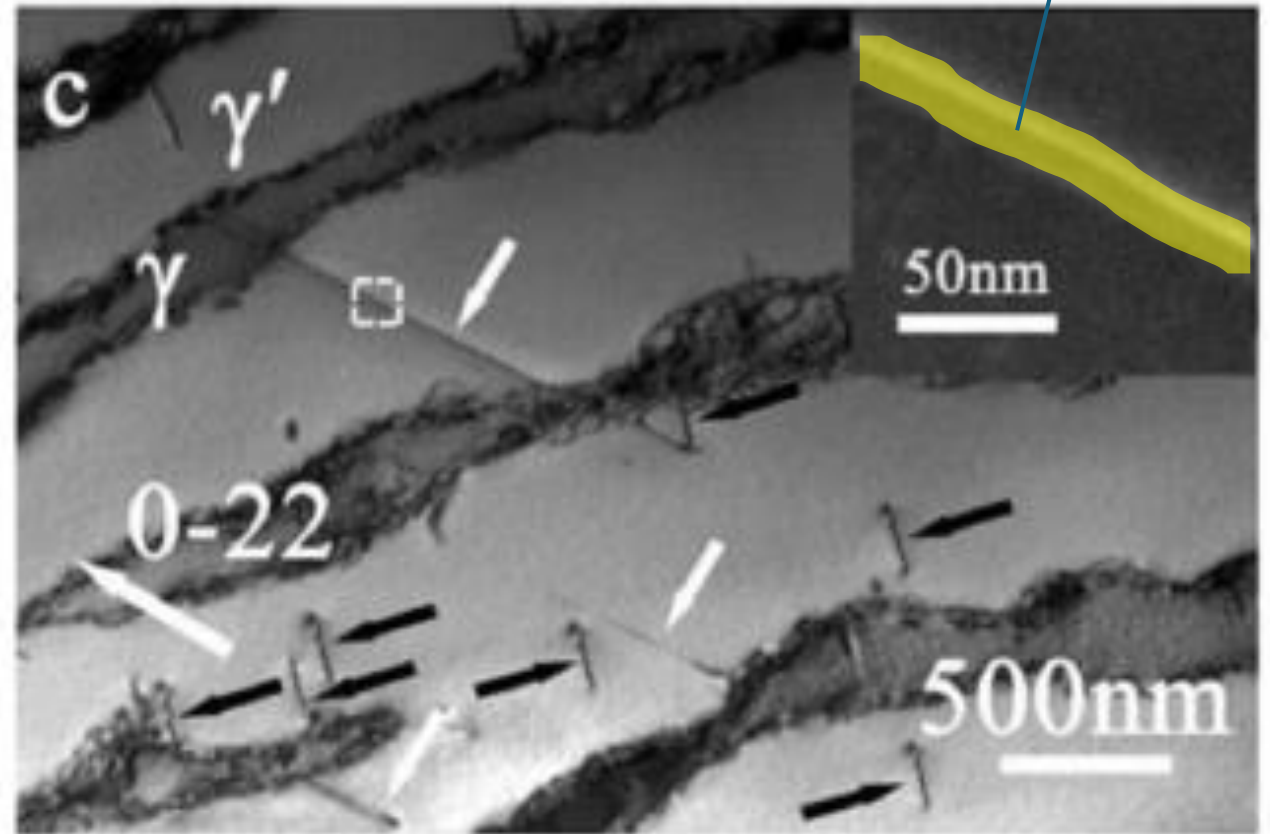
Forming an APB within an existing γ' crystal is the only way dislocations have, to pass through this phase.

APB = ANTI-PHASE BOUNDARIES

This is the reason why interfacial coherency enhances creep strength!

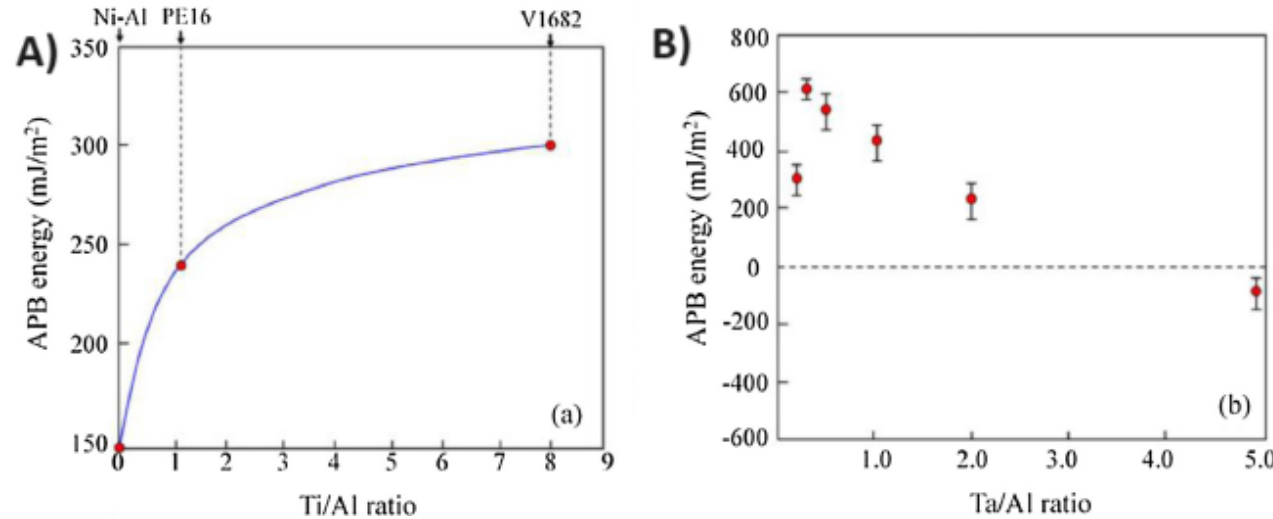
Schematic illustrations of dislocation movements in the γ channels

Diffusion of atoms and vacancies!!

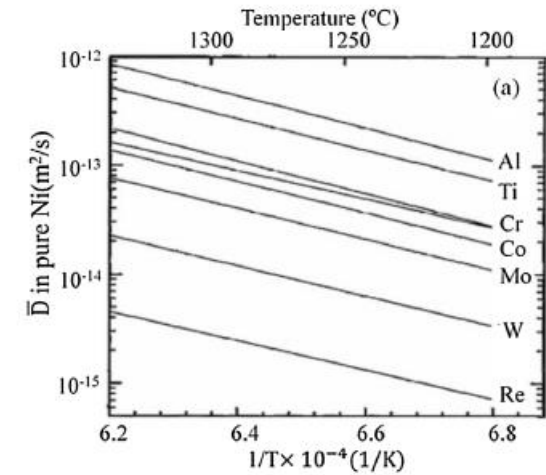


Illustrations of dislocation motion in γ' phase with forming APBs.

In light of the key factors discussed, which specific alloying elements are crucial for achieving the exceptional strength exhibited by this superalloy?



Effects of Ti and Ta on APB energy of Ni-SXs. The effects of Ti/Al ratios (a) and Ta/Al ratios (b) on APB energy in Ni-Al-X systems



Diffusion coefficients of some strengthening elements in Ni

$$r_{\text{Ti}}, r_{\text{Ta}} > r_{\text{Co}}, r_{\text{Cr}}, \text{etc.} \longrightarrow a_{\gamma'} > a_{\gamma} \longrightarrow \delta > 0$$

Why Al, Ti, Ta in particular? Larger than Ni.
It allows to have L1₂ structure (ordered)

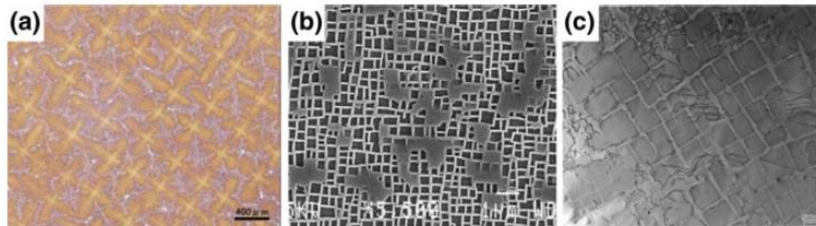
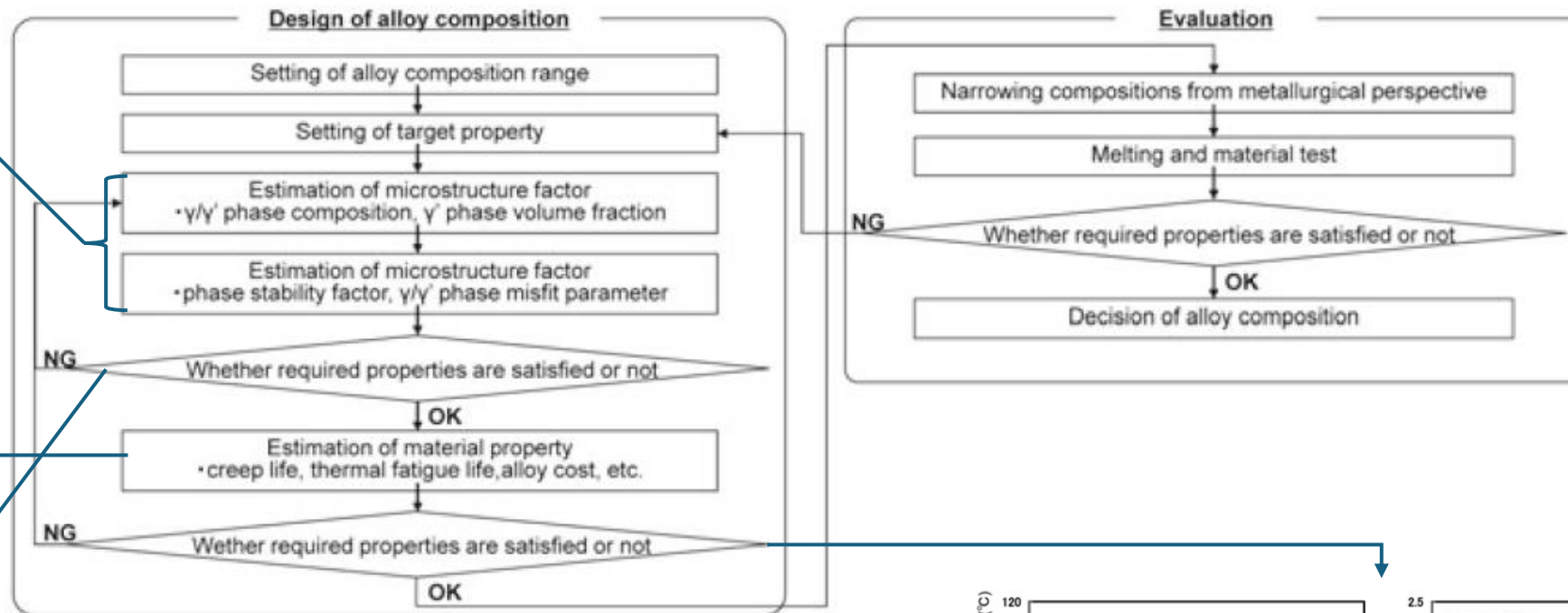
γ phase is a solid solution

High %Co reduces the energy of dislocations in γ phase

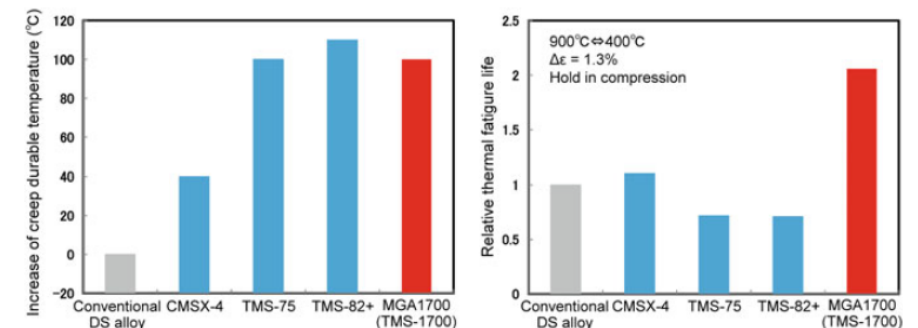
MGA1700: the result of joint research between NIMS and MHI

NIMS = National Institute for Materials Science

NIMS alloy design program

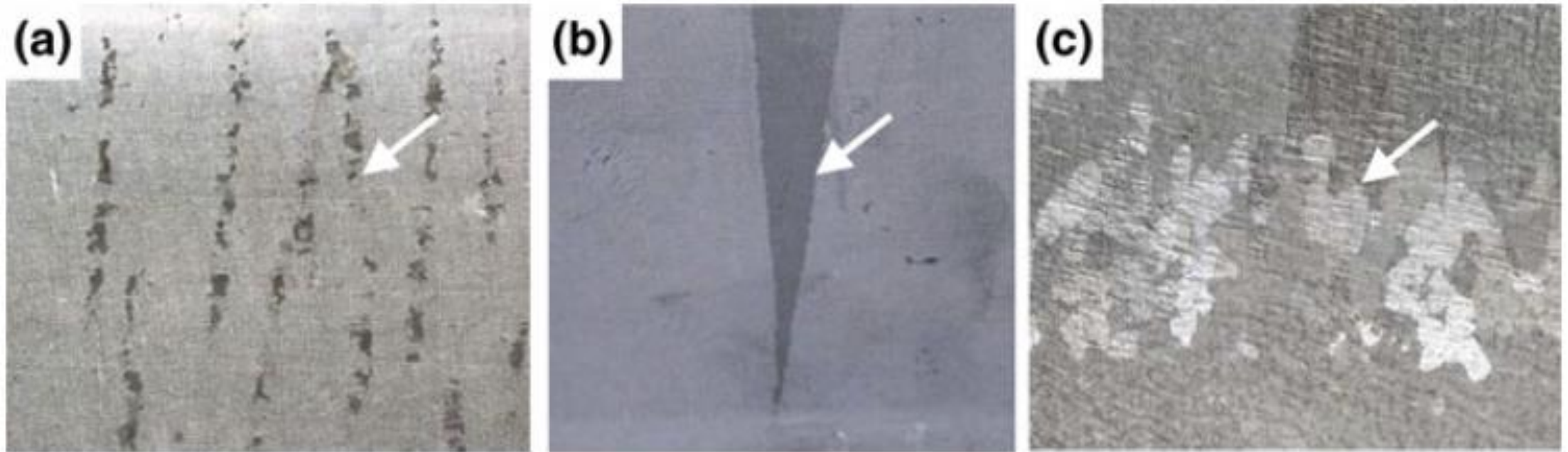


200 mm x 200 mm x 20 mm specimens



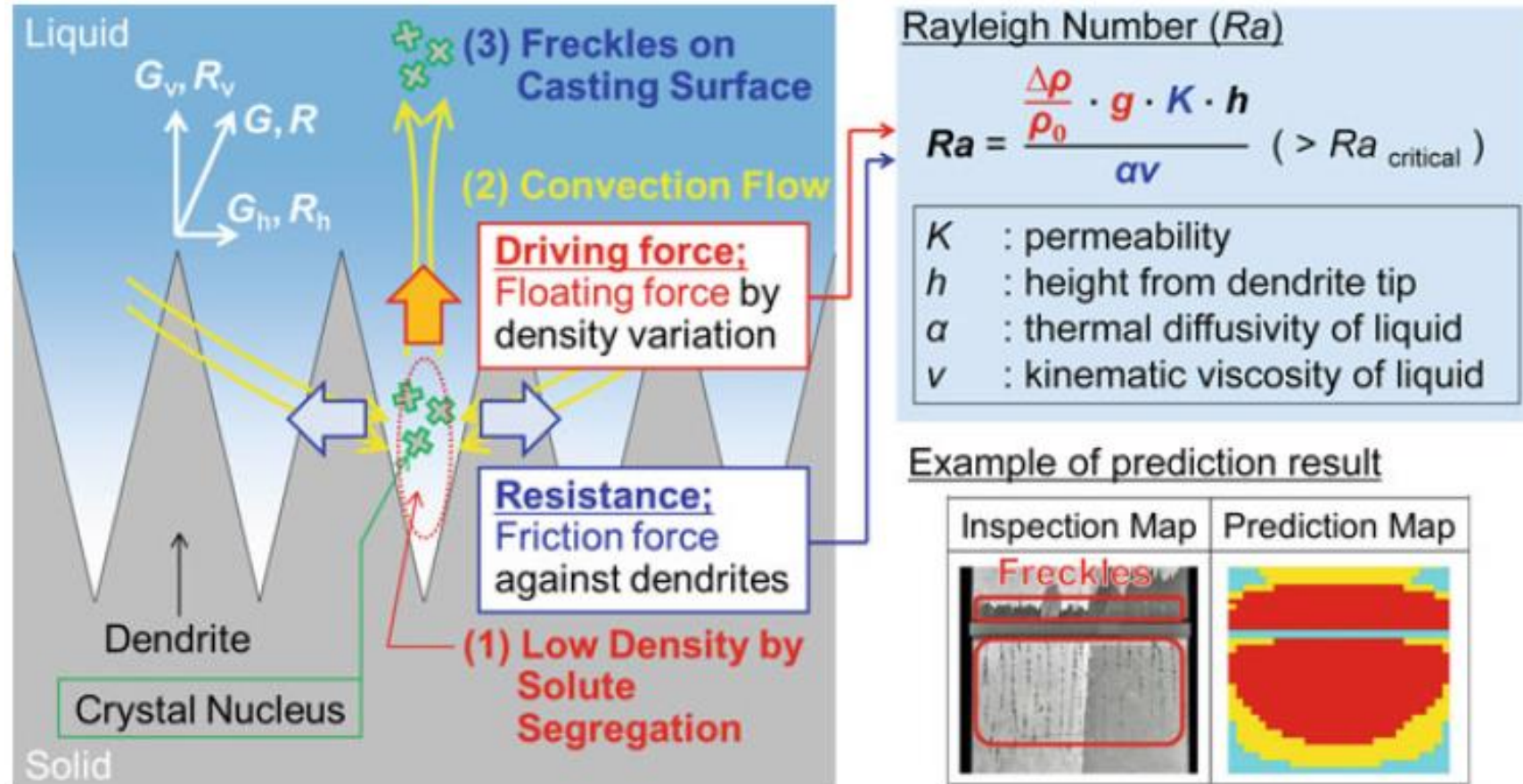
Results of testing on different composition

Typical casting defects in the SC superalloy



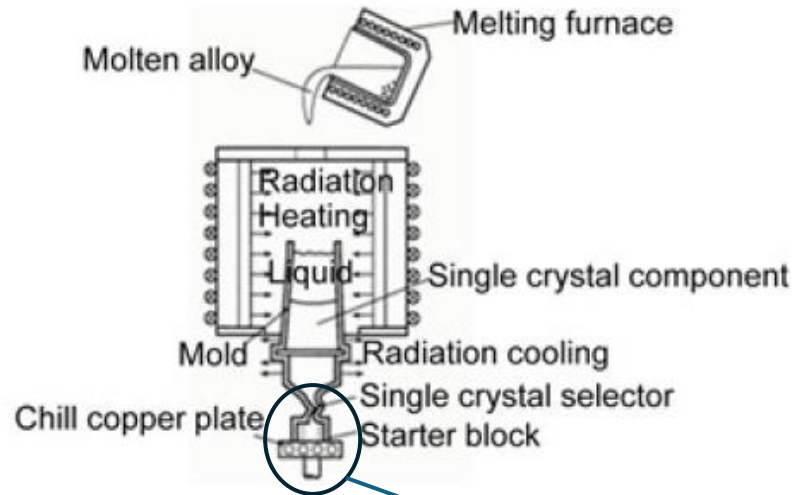
a) freckle, **b)** stray grain, and **c)** re-crystallization

Freckle generation

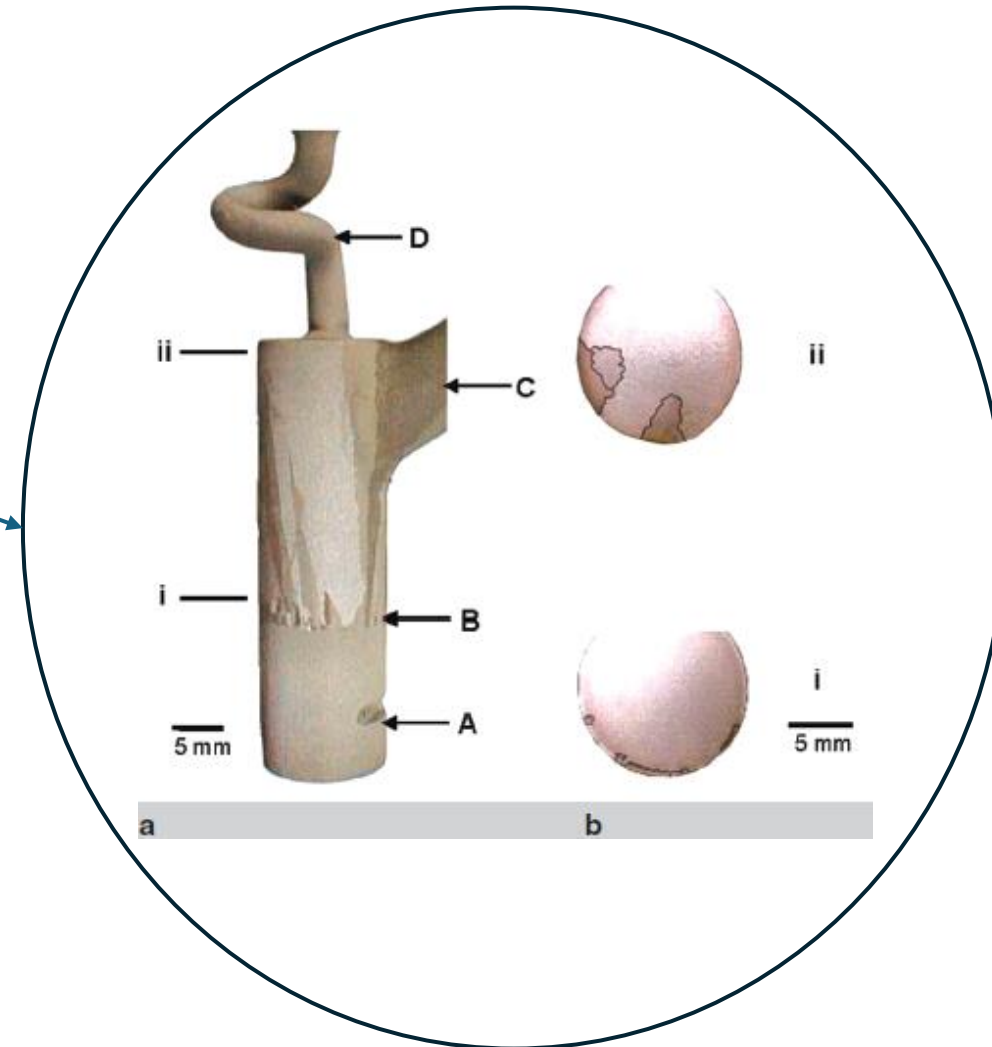


Basic mechanism and an example of prediction result of freckle generation

Casting process and stray grain

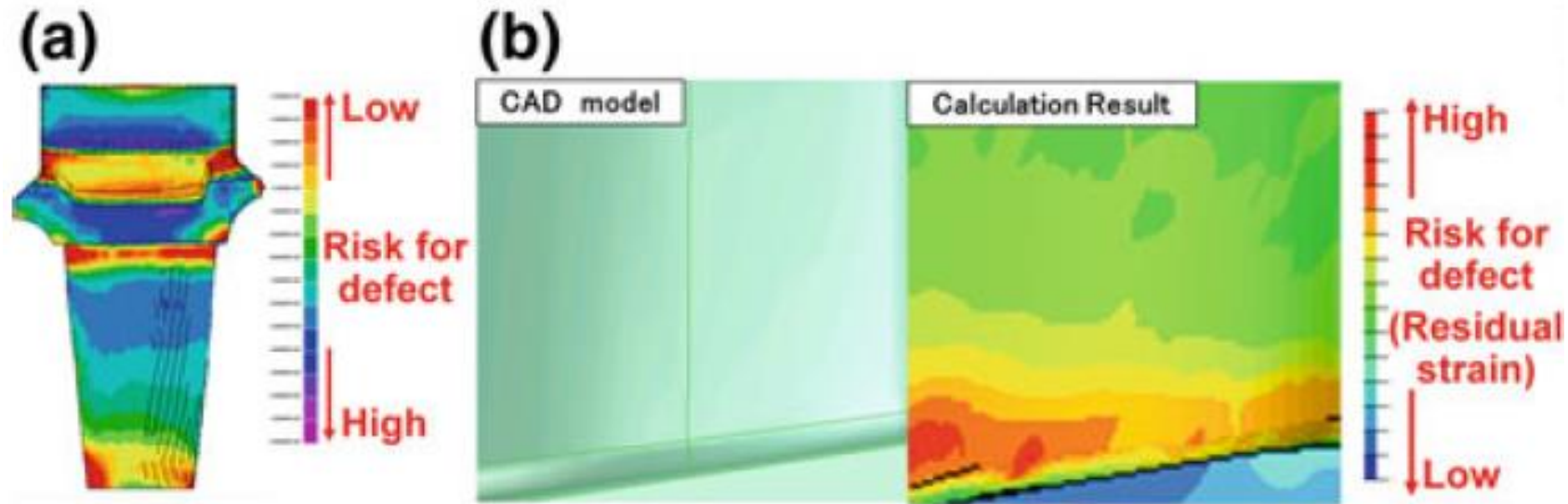


The seed and helical selector section of a turbine blade casting macro etched to highlight the stray grains



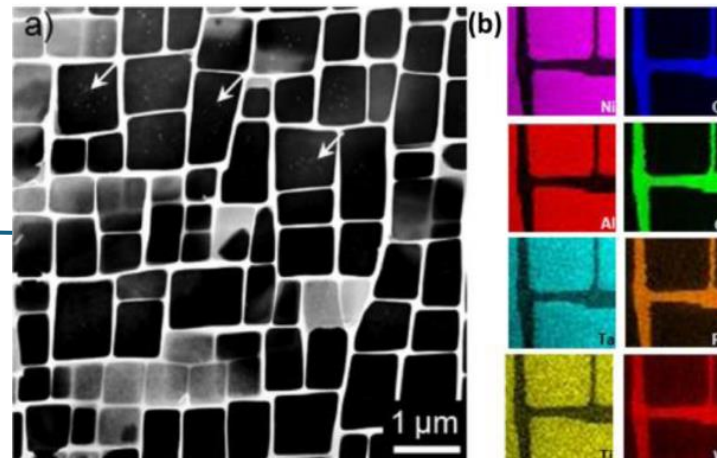
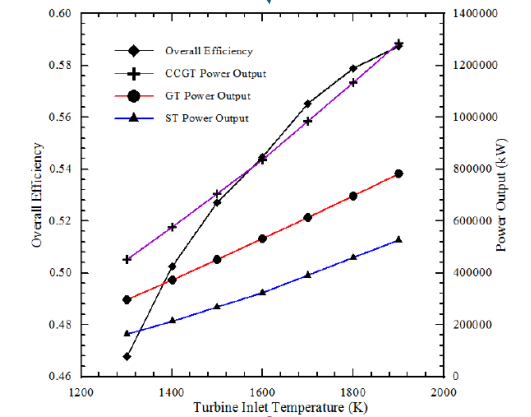
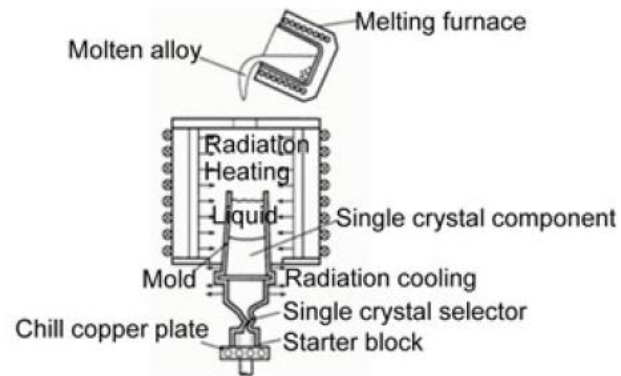
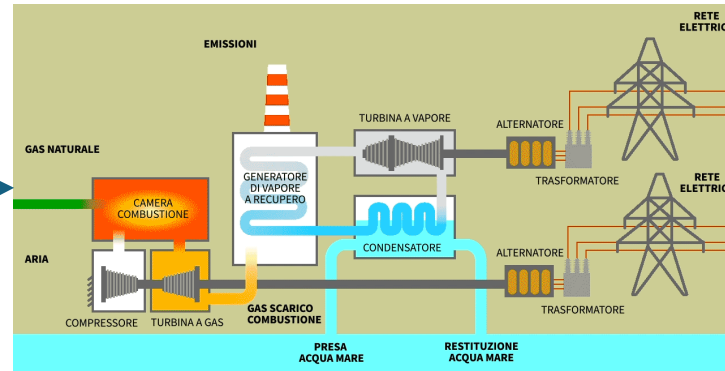
Solidification speed and thermal gradient to predict stray grain!

Stray grain and re-crystallization



Predicted results of **a)** stray grain, **b)** re-crystallization

Conclusion: multidisciplinary



Grazie dell'attenzione!